

COMBINATORICS OF COXETER GRAPHS

We start with the notion of level for a Coxeter graph.

Definition 1 (Level). *Let Γ be the Coxeter graph of an (irreducible) Coxeter system. Its level is the smallest non-negative integer r such that every choice of r vertices in Γ , when removed, leaves a (possibly disconnected) Coxeter graph of level 0 (finite or affine type).*

We denote level of a graph Γ by $L(\Gamma)$.

It'll be useful to define a set of all possible values of level of a Coxeter graph with n nodes.

$$L_n = \{ L(\Gamma) \mid \Gamma \text{ is a Coxeter graph with } n \text{ nodes} \}.$$

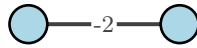
We can see that $n - 1$ is an upper bound for L_n .

However we also have the following,

Lemma 1. *$n - 1$ is a maximum for L_n .*

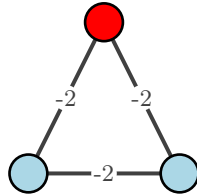
Proof. From the corollary above we know that $n-1$ is an upper bound for L_n . Hence, it suffices to show that $\exists$ a Coxeter graph with n nodes, say H , such that $L(H) = n - 1$.

Consider the following graph H^2 :

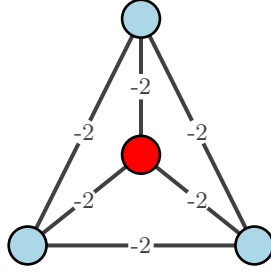


Notice that $L(H^2) = 1$ since $L(H^2) > 0$ and $L(H^2) \leq 1$.

Finally consider the following construction with H^2 : After adding a node we get H^3 , where $L(H^3) = 2$:



H^4 , where $L(H^4) = 3$



and so on....

We claim that $L(H^n) = n - 1$.

Therefore, a complete graph of n nodes with -2 on all edges gives us what we need.

Note that by symmetry on removing any node of H^n we get the graph H^{n-1} . In fact, on removing any choice of $n-2$ nodes we get H^2 . Hence, $L(H^n) > n-2$ and we also have $L(H^n) \leq n-1$. And so, $L(H^n) = n-1$ $\square$

We now define the natural partial order on the set of all Coxeter systems.

Definition 2. Let L be the set of all Coxeter graphs, then for $(W, S), (W', S') \in L$ with D, D' their associated Coxeter graphs.

$$D \leq D'$$

if D' is obtained from D by:

- (1) Adding exactly one more vertex and exactly one edge with the new vertex of weight 3 or
- (2) Increasing the weight of any of the edges by exactly 1 or adding an edge (of weight 3) for any two already existing vertices.

Definition 3. • If D' is obtained by (1) then we say D' is a **node-upper cover** of D , or vice-versa that D is a **node-lower cover** of D'

- Similarly, D' is a **label-upper cover** of D , or D is a **node-lower cover** of D' if D' is obtained by (2).

Lemma 2. Let Γ be a Coxeter graph, then for any $\Theta \leq \Gamma$, $Lvl(\Theta) \leq Lvl(\Gamma)$. (You can't have higher level graphs as a subgraph of lower level graphs)

Proof. If Θ is a label-lower cover of Γ then on removing r nodes from Γ say you get Γ' and Θ' from Θ , then clearly, $\Theta' \leq \Gamma'$ Since level of Γ' is 0, any lower cover of a level 0 graph is level 0 as well. This proves

that $Lvl(\Theta) \leq r$. Moreover, if Θ is a node-lower cover of Γ , then there exists a node on Γ , which on removing, gives Θ , let's call this node p . Say you remove r nodes from Γ to get Γ' , if you removed p , then $\Gamma' = \Theta'$ where Θ' is obtained by removing the remaining $r-1$ nodes. Since $Lvl(\Gamma') = 0 \implies Lvl(\Theta') = 0$. This actually proves that in this case $Lvl(\Theta) \leq r - 1$. $\square$

Theorem 1. (All Coxeter graphs of level r "come" from Coxeter graphs of the same level or of one level lower)

Let Γ be an arbitrary Coxeter graph such that $Lvl(\Gamma) = r > 1$ then there exists a coxeter graph, Θ , such that $\Theta \leq \Gamma$ and $r - 1 \leq Lvl(\Theta) \leq r$.

Proof. Say by contradiction, that all lower covers of Γ are of level $r - 2$ or lower. Consider the act of removing any one of the nodes from Γ to obtain Γ' . Since the action of removing any one of the nodes is stronger than (1) i.e there exists $\Theta \leq \Gamma$ such that $\Gamma' \leq \dots \leq \Theta \leq \Gamma$. We have that $Lvl(\Gamma') \leq Lvl(\Theta) \leq r - 2$, and so removing $r - 2$ nodes from Γ' gives us a level 0 graph. This "action" covers every way of removing $r - 1$ nodes from Γ and we see that we get a level 0 graph every time $\implies Lvl(\Gamma) \leq r - 1$ which contradicts our assumption. $\square$

Lemma 3. Let Γ be a Coxeter graph of $Lvl(\Gamma) = r > 1$ then on removing any one of the nodes from Γ one gets a Coxeter graph of level less than or equal to $r - 1$.

Proof. Say by contradiction, there exists a node which on removing from Γ gives Γ' such that $Lvl(\Gamma') = r$ ($Lvl(\Gamma)$ cannot be greater than r). Since $Lvl(\Gamma') = r \implies Lvl(\Gamma') > r - 1$ hence there exists a choice of $r - 1$ nodes which, on removing, gives a graph of level greater than 0 $\implies Lvl(\Gamma) > r$, since we have a choice of r nodes, which on removing, doesn't give a level 0 Coxeter graph, a contradiction. $\square$

Corollary 1. (1) If Θ is a any node-lower cover of Γ , then $Lvl(\Theta) \leq Lvl(\Gamma) - 1$.
 (2) If $\Theta \leq \Gamma$ such that $Lvl(\Gamma) = Lvl(\Theta)$, then Γ is a label-lower cover of Γ

It'll be useful to define the set of all Coxeter graphs of level $r \geq 0$:

$$L(r) := \{\Gamma : \text{A Coxeter graph such that } Lvl(\Gamma) = r\}$$

Definition 4. (Minimal graphs of a level) Let $\Gamma \in L(r)$, Then for $r > 1$, Γ is a minimal Coxeter graph of level r if all lower covers of Γ are of level less than or equal to $r - 1$. Denote all minimal level graphs of a level by $M(r)$

Definition 5. (*Boundary graphs of a level*) Let $\Gamma \in L(r)$, Then for $r \geq 0$, Γ is a boundary Coxeter graph of level r if all upper covers of Γ are of level more than or equal to $r + 1$.

Theorem 2. All minimal graphs of a level, say $r > 1$, are either

- (1) Node upper covers of minimal graphs of level $r-1$.
- (2) Label upper covers of boundary graphs of level $r-1$.

Algorithm 1. (*Label Chain*). For any minimal Coxeter graph $M \in L(r)$, get all label-upper covers of M that are of level r , call it $LU_r(M)$. For each $N \in LU_r(M)$, get all label-upper covers of N that are of level r . Repeat the process until no upper covers are of level r are remaining. Denote the whole "chain" thus obtained (including M) with $LC(M)$.

Theorem 3. The set of all Coxeter graphs of level r , $L(r)$ is a disjoint union:

$$L(r) = \coprod_{M \in M(r)} LC(M)$$